

1. Given the curve $\mathcal{C}$ described by the vector function $\vec{r}(t) = (2 \cos^2 t - t, 2t - \sin t, \sin(2t) + 2)$ with $-\pi/4 \leq t \leq \pi$.

- (a) Find the unit tangent vector $\vec{T}$ at the point $(2, 0, 2)$.

We first seek t such that $\vec{r}(t) = (2, 0, 2)$. That is,

$$\begin{cases} 2 \cos^2 t - 2 = 2, \\ 2t - \sin t = 0, \\ \sin(2t) + 2 = 2. \end{cases}$$

From the last equation, we find that

$$\sin(2t) = 0 \quad \Rightarrow \quad 2t = 0, \pi, 2\pi \quad \Rightarrow \quad t = 0, \frac{\pi}{2}, \pi.$$

However, only $\vec{r}(0) = (2, 0, 2)$.

Now, we have been asked to find $\vec{T}(0) = \frac{\vec{r}'(0)}{|\vec{r}'(0)|}$. Notice that

$$\vec{r}'(t) = (-4 \cos t \sin t - 1, 2 - \cos t, 2 \cos(2t)) \quad \Rightarrow \quad \vec{r}'(0) = (-1, 1, 2).$$

Thus,

$$\vec{T}(0) = \frac{1}{\sqrt{6}}(-1, 1, 2).$$

- (b) Find the curvature κ at the point in (a);

$$\vec{r}''(t) = (-4 \cos(2t), \sin t, -4 \sin(2t)) \quad \Rightarrow \quad \vec{r}''(0) = (-4, 0, 0).$$

$$\vec{r}'(0) \times \vec{r}''(0) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 1 & 2 \\ -4 & 0 & 0 \end{vmatrix} = (0, -8, 4)$$

Therefore, the curvature at the point $(2, 0, 2)$ is

$$\kappa(0) = \frac{|\vec{r}'(0) \times \vec{r}''(0)|}{|\vec{r}'(0)|^3} = \frac{2}{3} \sqrt{\frac{5}{6}}$$

- (c) Knowing that $\vec{B}(t) = \frac{\vec{r}'(t) \times \vec{r}''(t)}{|\vec{r}'(t) \times \vec{r}''(t)|}$, find the principal unit normal vector $\vec{N}$ to the curve $\mathcal{C}$ at the point in (a).

We first find

$$\vec{B}(0) = \frac{\vec{r}'(0) \times \vec{r}''(0)}{|\vec{r}'(0) \times \vec{r}''(0)|} = \frac{1}{\sqrt{5}}(0, -2, 1).$$

Now, the principal unit normal vector is

$$\vec{N}(0) = \underbrace{\vec{B}(0) \times \vec{T}(0)}_{\text{The order}} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{vmatrix} = \frac{1}{\sqrt{30}}(-5, -1, -2)$$

(d) Find the cartesian equation of the rectifying plane to the curve $\mathcal{C}$ at the point in (a).

By definition, the rectifying plane is determined by the vectors $\vec{B}(0)$ and $\vec{T}(0)$ and hence, has normal vector $\vec{N}(0)$. Thus, its cartesian equation is

$$(x - 2, y, z - 2) \cdot \frac{1}{\sqrt{30}}(-5, -1, -2) = 0 \quad \Leftrightarrow \quad -5x - y - 2z = -14.$$

2. Let the variable force $\vec{f}(t) = (\cos t, 0, 1)$ move an object of mass $m = 3 \text{ kg}$ along a curve $\mathcal{C}$ from an initial position $\vec{r}(0) = (1, 2, 3)$ with an initial velocity $\vec{r}'(0) = (-1, -3, 0)$. Find the position $\vec{r}(t)$ of the object at every instant t . (Hint: Newton's second law of motion states that $\vec{f}(t) = m\vec{r}''(t)$).

From $\vec{f}(t) = m\vec{r}''(t)$, we get that $\vec{r}''(t) = \frac{1}{3}(\cos t, 0, 1)$, then integrate twice to find $\vec{r}(t)$. That is,

$$\vec{r}'(t) = \vec{r}'(0) + \int_0^t \vec{r}''(s)ds = (-1, -3, 0) + \frac{1}{3}(\sin t, 0, t) = \left(-1 + \frac{1}{3}\sin t, -3, \frac{t}{3}\right)$$

and

$$\begin{aligned} \vec{r}(t) &= \vec{r}(0) + \int_0^t \vec{r}'(s)ds = (1, 2, 3) + \left(-t - \frac{1}{3}\cos t, -3t, \frac{1}{6}t^2\right) \\ &= \left(\frac{4}{3} - t - \frac{1}{3}\cos t, 2 - 3t, 3 + \frac{1}{6}t^2\right). \end{aligned}$$

3. The particles collide if they are at the same position at the same instant. So we solve t such that $\vec{r}_1(t) = \vec{r}_2(t)$ which implies $\cos t = \sin t = 0$ which is impossible because $\cos^2 t + \sin^2 t = 1$.

To find the intersection(s) of their paths we need solve t and s such that $\vec{r}_1(t) = \vec{r}_2(s)$. That is,

$$\boxed{0 = \cos t, \quad \cos s = \sin t, \quad \sin s = 0}.$$

The first and third equations imply that $\sin t = \pm 1$ and $\cos s = \pm 1$ respectively. Hence, taking into account the second equation we get $\cos s = \sin t = -1$ or $\cos s = \sin t = 1$. Therefore, the two paths intersect at the points $(0, -1, 0)$ and $(0, 1, 0)$.

4. (a) The domain of the function $f_1(x, y) := \frac{2}{\ln(1 - x^2 - y^2)}$ is

$$D_1 := \{(x, y) \in \mathbb{R}^2: x^2 + y^2 < 1 \text{ with } (x, y) \neq (0, 0)\}.$$

- (b) The domain of $f_2(x, y) = \sqrt{y^2 - x^4}$ is $D_2 := \{(x, y) \in \mathbb{R}^2: y \geq x^2 \text{ or } y \leq -x^2\}$;

- (c) The domain of $f_3(x, y) = \ln(1 - x^2) + \ln(1 - y^2)$ is $D_3 := (-1, 1) \times (-1, 1)$;

- (d) The domain of $f_4(x, y) = \ln\left(\frac{1 - x^2}{1 - y^2}\right)$ is

$$D_4 := \{(x, y) \in \mathbb{R}^2: 1 - x^2 > 0 \text{ and } 1 - y^2 > 0\} \cup \{(x, y) \in \mathbb{R}^2: 1 - x^2 < 0 \text{ and } 1 - y^2 < 0\}.$$

Sketch these domains in the plane.